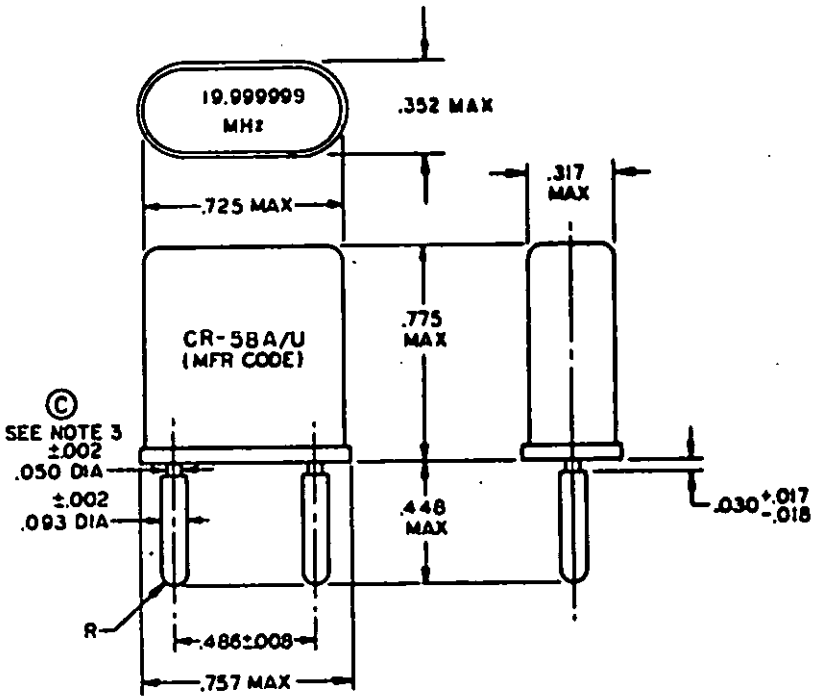
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-58A/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the crystal unit described herein shall consist of this document and the latest issue of Specification MIL-C-3098.

Pertinent characteristics: 0.8 to 20 MHz; fundamental; noncontrolled; antiresonance.



INCHES	MM	INCHES	MM	INCHES	MM
.002	.05	.050	1.27	.486	12.34
.008	.20	.093	2.36	.725	18.42
.017	.43	.317	8.05	.757	19.23
.018	.46	.352	8.94	.775	19.69
.030	.76	.448	11.38		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
© 3. The pin undercut may be omitted.

© denotes change

FSC 5955

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TABLE I. Requirements.

Requirements	*Specified value or condition*
Holder	HC-17/U (MIL-H-10056/8)
Dimensions, marking, and configuration	See figure
Frequency range	0.8 to 20 MHz, inclusive
Capacitance, shunt	7 *pF, maximum*
Frequency tolerance (operating temperature range)	±50 PPM
Equivalent resistance	See table III
Mode of oscillation	Fundamental
Antiresonance, load capacitance	32.0 ± 0.5 *pf*
Operating temperature range	-55° to +105° C inclusive
Test set, calibration values, and rated drive level	See table II
Shock (specified pulse):	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	Below 2 MHz, ±15 percent 2 MHz and above, ±10 percent
Vibration: 1/	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	Below 2 MHz, ±15 percent 2 MHz and above, ±10 percent
Temperature run:	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	Below 2 MHz, ±15 percent 2 MHz and above, ±10 percent
Aging:	
Frequency change permitted	5 PPM

1/ Method 201 of MIL-STD-202.

TABLE II. Test set, calibration values, and rated drive level.

Government reference standard test set	Band of specified frequencies inclusive	Calibration values			Rated drive level
		Resistance	Crystal current	Resistor voltage drop	
	MHz	Ohms	mA	Volt	mW
TS-330/TSM	0.8 to 1.5	100	10	---	10.0 ± 2.0
	1.5+ to 2.25	50	15	---	
	2.25+ to 3.4	40	15	---	
	3.4+ to 5.1	25	20	---	
	5.1+ to 7.5	14	25	---	
	7.5+ to 10	11	30	---	
	10+ to 15	13	20	---	5.0 ± 1.0
TS-683/TSM	15+ to 20	10	---	0.22	5.0 ± 1.0

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TABLE III. Equivalent resistance.

Frequency range, inclusive	Maximum resistance
MHz	Ohms
0.8 to 0.85	630
0.85+ to 0.9	600
0.9+ to 1	580
1+ to 1.12	540
1.12+ to 1.25	490
1.25+ to 1.37	450
1.37+ to 1.5	410
1.5+ to 1.62	380
1.62+ to 1.75	330
1.75+ to 1.87	300
1.87+ to 2	290
2+ to 2.12	270
2.12+ to 2.25	250
2.25+ to 2.6	200
2.6+ to 3	150
3+ to 3.4	110
3.4+ to 3.75	90
3.75+ to 4	75
4+ to 5	60
5+ to 7	35
7+ to 10	24
10+ to 15	22
15+ to 20	20

NOTES:
1. Socket. This unit has oversize pins and does not use same socket as preferred types of crystal units.
2. This crystal-unit type is to be used for special applications and for replacement purposes only. Specific approval of the Government is required prior to each intended use. See MIL-STD-683.

Custodians:
Army - EL
Navy - EC
Air Force - 84

Review activities:
Army - EL, MI, MU, SG, SM
Navy - EC
Air Force - 11, 85
DSA - ES

User activities:
Army -
Navy - AS, MC, SH
Air Force - 17, 19

Preparing activity:
Army - EL

Agent:
DSA - ES

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